

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

KLE TECHNOLOGICAL UNIVERSITY, HUBBALLI, INDIA

Course Project Report

On

**Aviation Data Warehouse for Flight Delay Analytics**

submitted in partial fulfilment of the requirements for the award of the degree of

Bachelor of Engineering

IN

COMPUTER SCIENCE AND ENGINEERING

*Submitted By*

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# 1. Introduction

## 1.1 Preamble

The modern aviation industry generates massive volumes of operational data daily, making it critical to have efficient systems to track flight schedules, airline performance, and airport traffic. This project models an Aviation Data Warehouse that organizes historical commercial flight data to support operational analytics and delay tracking. Using Informatica Intelligent Data Management Cloud (IDMC), the system enables streamlined data extraction, cleansing, transformation, and loading (ETL) to power business intelligence.

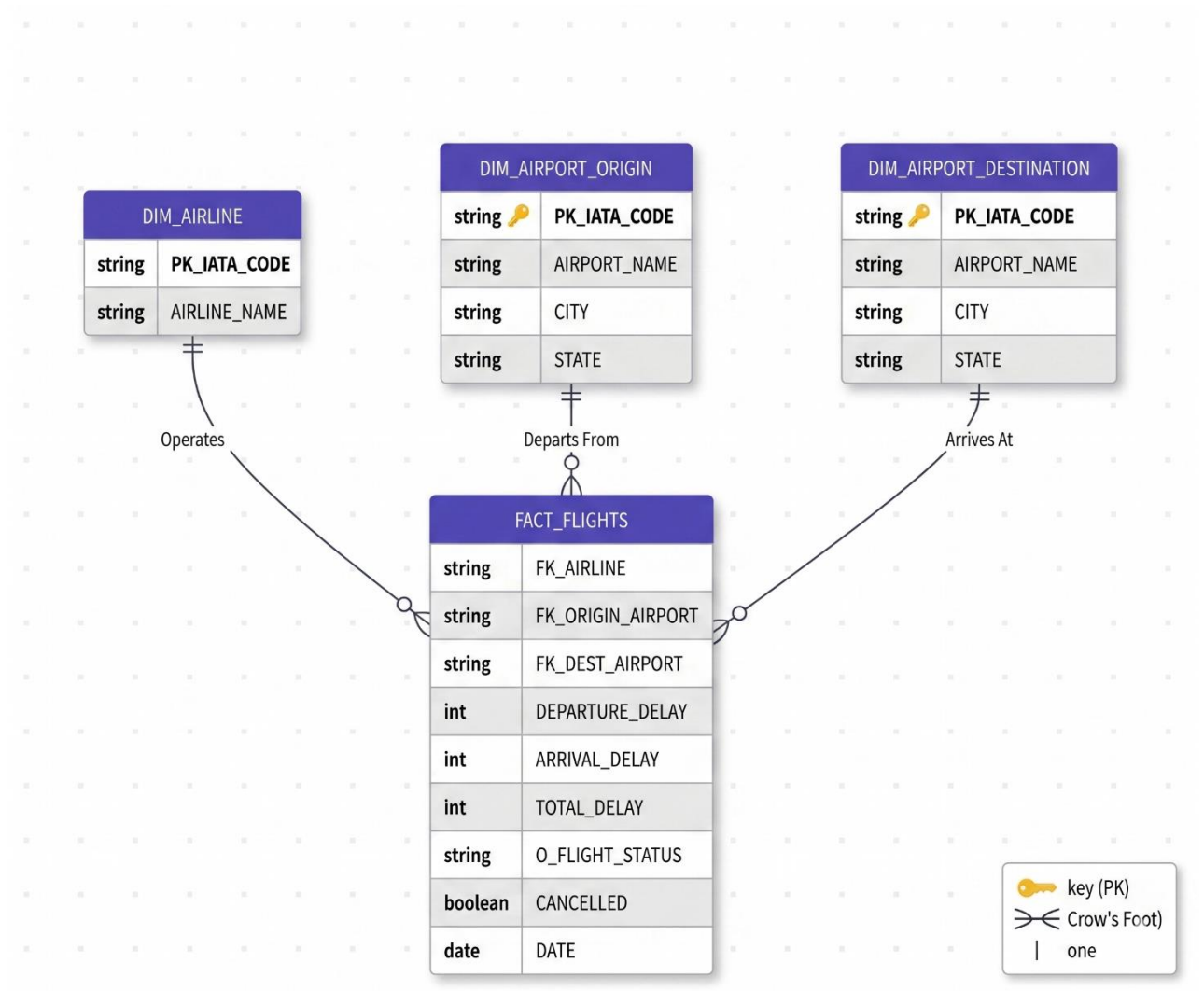
## 1.2 Problem Definition

Raw aviation data is often distributed across massive, messy flat files (CSVs) containing millions of records. Aviation authorities and data analysts struggle to generate insights because the raw data contains duplicates, missing values, and uncalculated metrics (like total delay times). Furthermore, there is a lack of structured relationships between the raw flight logs and the reference tables for airlines and airports, making it difficult to generate accurate, fast performance analytics.

## 1.3 Objectives

- To design and implement a relational Star Schema data model for aviation analytics.
- To utilize Informatica IDMC to build enterprise-grade ETL pipelines.
- To cleanse and deduplicate reference data to build Dimension tables (Airports and Airlines).
- To perform real-time data calculations and route data conditionally into appropriate Fact tables based on flight status (Delayed vs. On-Time).
- To aggregate millions of rows of data to calculate high-level business metrics, such as average delay times per airline.

## 2. ER Diagram



# 3. Dataset Description

This project utilizes a large-scale commercial aviation dataset (sourced from Kaggle). The original raw data is distributed across three separate flat files (CSVs) which act as the foundational sources for the ETL pipelines.

## 2.1 flights.csv

**Description:** This is the massive core dataset (approximately 578 MB) containing millions of individual flight records. It serves as the raw, unstructured source for the central Fact table, containing historical logs of scheduled flights, actual departure/arrival times, and delay metrics.

### Key Columns:

- **YEAR, MONTH, DAY, DAY\_OF\_WEEK:** Date and time parameters indicating exactly when the scheduled flight took place.
- **AIRLINE:** The raw 2-letter identifier for the airline operating the flight.
- **ORIGIN\_AIRPORT:** The raw 3-letter IATA code indicating where the flight took off from.
- **DESTINATION\_AIRPORT:** The raw 3-letter IATA code indicating where the flight landed.
- **DEPARTURE\_DELAY:** Integer representing the number of minutes the flight was delayed upon take-off. (Negative numbers indicate early departures) .
- **ARRIVAL\_DELAY:** Integer representing the number of minutes the flight was delayed upon landing.
- **CANCELLED:** A binary indicator (1 or 0) showing whether the flight was cancelled before take-off.

## 2.2 airports.csv

**Description:** A master reference file (approximately 24 KB) containing geographical and naming data for commercial airports. This file requires cleansing and deduplication before being loaded as a formal Dimension table.

**Key Columns:**

- **IATA\_CODE:** The unique 3-letter code assigned to the airport (e.g., JFK, LAX, ORD).
- **AIRPORT:** The official, full-text name of the airport facility.
- **CITY:** The city where the airport is geographically located.
- **STATE:** The state or region where the airport is located.

## 2.3 airlines.csv

**Description:** A lightweight master reference file (approximately 1 KB) that maps the abbreviated airline codes used in the flights.csv file to their full corporate company names.

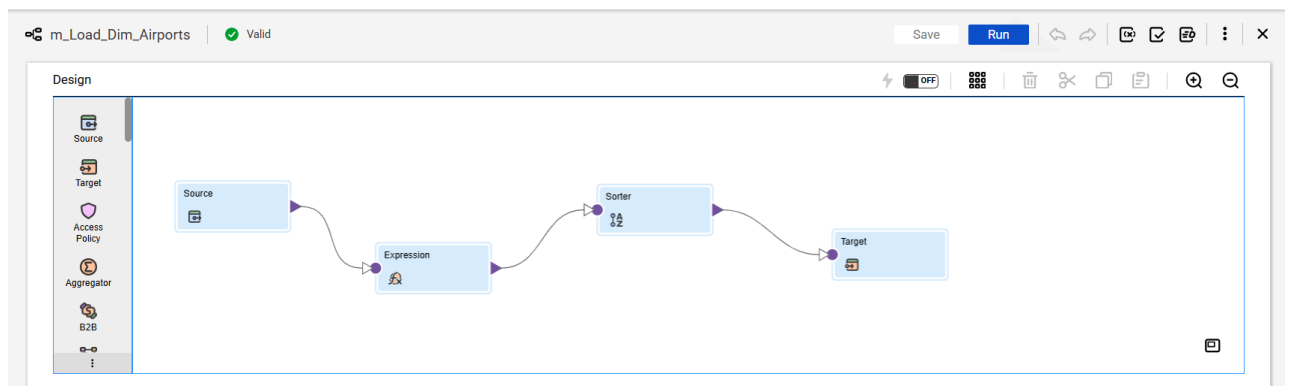
**Key Columns:**

- **IATA\_CODE:** The unique 2-letter identifier assigned to the airline (e.g., UA, DL, AA).
- **AIRLINE:** The official, full-text name of the commercial airline (e.g., United Air Lines Inc., Delta Air Lines, American Airlines).

# 4. Transformations With Scenarios

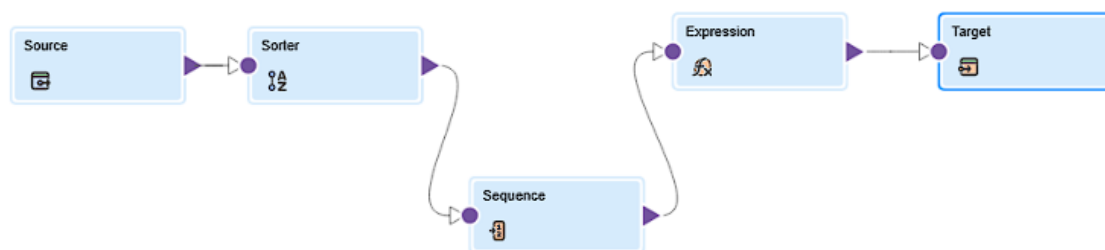
## Scenario 1: Cleansing and Deduplicating Airport Reference Data (m\_Load\_Dim\_Airports)

- **Business Need:** The raw list of airports is messy and contains duplicate entries. We need to clean it up before building our core reference tables.
- **Transformation Logic:** An **Expression** transformation is used to trim away unwanted, invisible spaces from the text using LTRIM(RTRIM()). The cleaned data then passes through a **Sorter** transformation with the 'Distinct' property enabled. This permanently deletes any duplicate records before saving the data to the final Dimension table.
- **Transformation:**



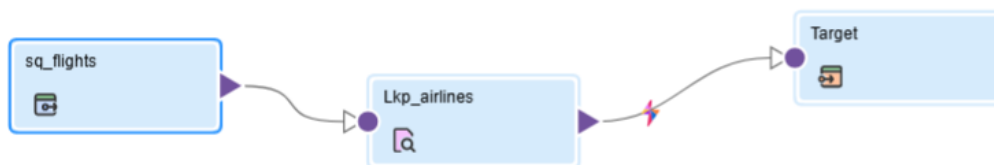
## Scenario 2: Dimension Surrogate Key Generation (m\_Load\_Dim\_Airlines\_Surrogate)

- **Business Need:** To make database searches faster, we need to replace text-based airline codes (like "DL") with simple, unique ID numbers (like 1, 2, 3...).
- **Transformation Logic:** The airline data is organized alphabetically using a **Sorter** transformation. A **Sequence Generator** is then used to count upwards from 1, assigning a unique number to each airline. Finally, an **Expression** merges this new ID into the final target table.
- **Transformation:**



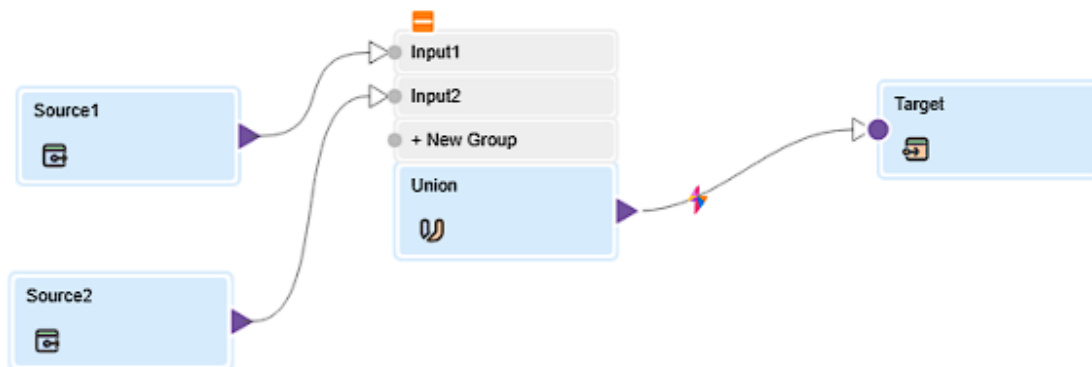
### Scenario 3: Master Data Translation and Naming Collision Resolution (m\_Master\_Data\_Translator)

- **Business Need:** Raw flight logs only show short, confusing airline codes. We need to translate these into full corporate names (e.g., "United Airlines") so business dashboards are easy to read.
- **Transformation Logic:** A **Lookup** transformation acts as a dictionary, cross-referencing the short codes against our clean airline file. To prevent column names from getting mixed up during the merge, we apply a bulk prefix (LKP\_) to the output ports, ensuring the translated names map perfectly to the target file.
- **Transformation:**



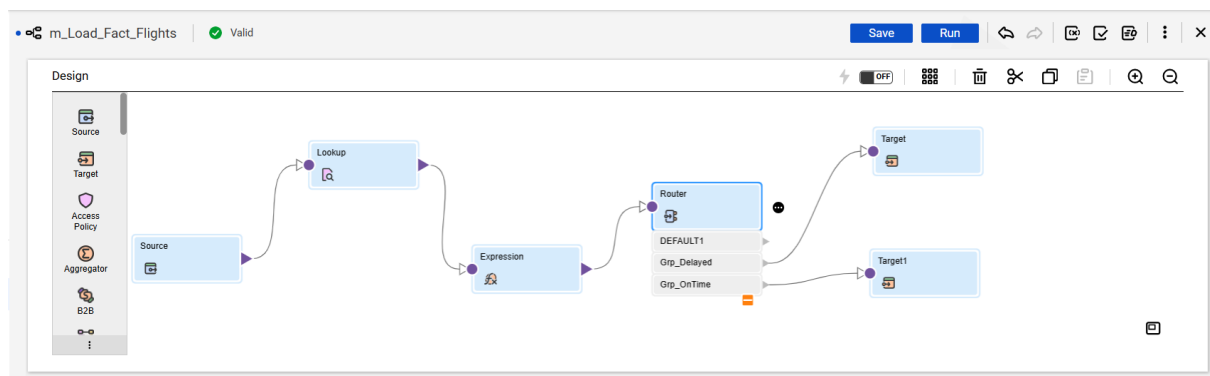
## Scenario 4: Merging Fragmented Data Streams (m\_Union\_Data\_Streams)

- **Business Need:** Flight data often arrives in separate pieces from different regional servers. We need to stack them together into one massive master list.
- **Transformation Logic:** Two separate **Source** transformations pull in the fragmented files. A **Union** transformation then acts as a funnel. By using 'Exact Match' mapping, it stacks the incoming data vertically, creating a single, unified pipeline.
- **Transformation:**



## Scenario 5: Conditional Routing and Fact Table Generation (m\_Load\_Fact\_Flights)

- **Business Need:** We need to calculate the total delay time for every flight and then split the data: sending delayed flights to one table and on-time flights to another.
- **Transformation Logic:** An **Expression** calculates the total delay by adding departure and arrival times. Then, a **Router** transformation acts as a traffic cop. It splits the data stream, sending flights delayed by more than 15 minutes down one path, and on-time flights down another.
- **Transformation:**



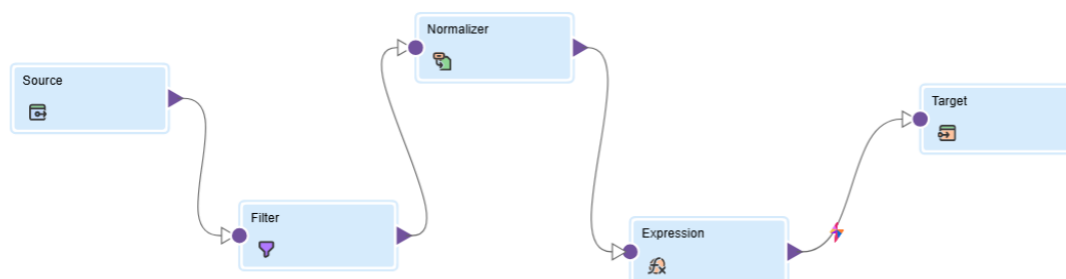
## Scenario 6: Categorical Derivations for Machine Learning (m\_Load\_Fact\_Flight\_Delays)

- **Business Need:** Data science AI models need simple text categories (like "Minor Delay" or "Severe Delay") instead of raw numbers (like 45 minutes).
- **Transformation Logic:** An **Expression** converts the text-based delay numbers into integers using TO\_INTEGER(). Then, it uses nested IIF() statements to evaluate the time and automatically assign the correct text label to a new category column.
- **Transformation:**



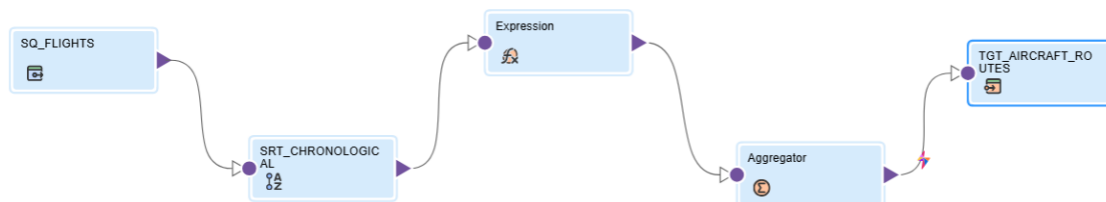
## Scenario 7: Root-Cause Unpivotter for Business Intelligence (m\_Normalize\_Flight\_Delays)

- **Business Need:** The data currently lists five different delay reasons in a wide layout (five columns). Charting tools like Tableau need this data squished into a tall layout (one column) to build charts properly.
- **Transformation Logic:** A **Filter** first drops all on-time flights to save processing power. A **Normalizer** transformation then collapses the 5 separate columns into a single column. Finally, an **Expression** uses a DECODE() function to translate the system's index numbers back into readable English text.
- **Transformation:**



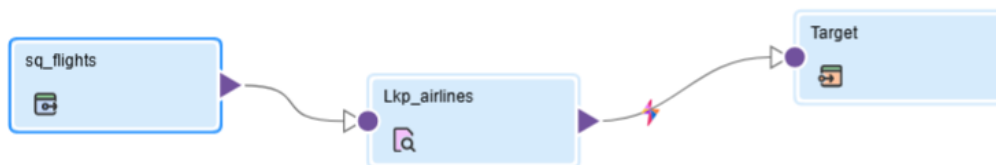
## Scenario 8: Historical Flight Path Tracer (m\_Flight\_Path\_Tracer)

- **Business Need:** The operations team wants to see the daily route of specific planes (e.g., JFK -> ORD -> LAX), but the raw data only shows disconnected flights. We need to stitch them together.
- **Transformation Logic:** A **Sorter** organizes the flights chronologically by the plane's tail number. An **Expression** uses variable ports to connect the origin and destination codes row-by-row into a growing string. Finally, an **Aggregator** groups the data by tail number to output only the final, completed route.
- **Transformation:**



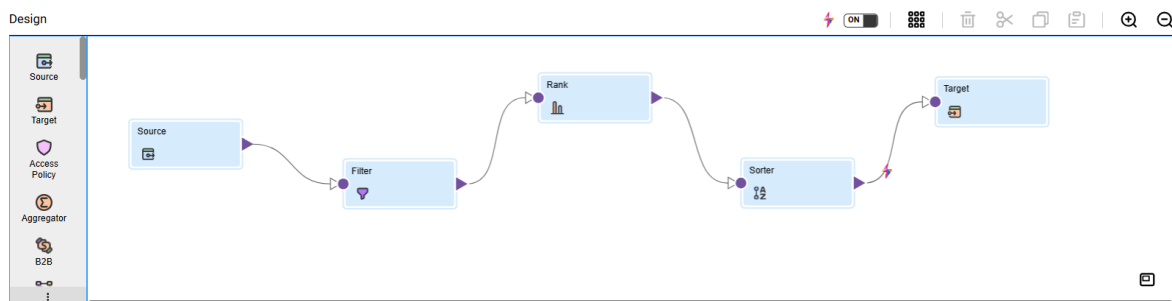
## Scenario 9: Isolating Top Outliers for Executive Audits (m\_Load\_Top10\_Longest\_Flights)

- **Business Need:** The operations team needs a quick daily list of the top 10 longest flights by distance to study fuel efficiency.
- **Transformation Logic:** An **Expression** ensures the distance data is treated as a number. Then, a **Rank** transformation searches the massive dataset, grabs only the top 10 highest values, and routes them directly to a clean target file.
- **Transformation:**



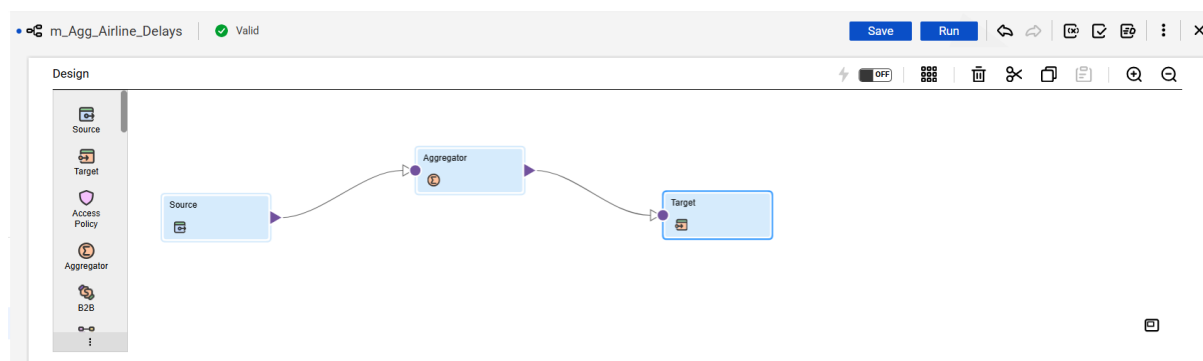
## Scenario 10: Isolating and Ordering Extreme Performance Outliers (m\_Rank\_Worst\_Delays)

- **Business Need:** Executives need to see the absolute worst 50 delayed flights to investigate operational failures, but we cannot let cancelled flights skew the report.
- **Transformation Logic:** A **Filter** transformation removes all cancelled flights. A **Rank** transformation then isolates the 50 highest departure delays. Finally, a **Sorter** organizes these 50 rows in strict descending order, from worst to least bad.
- **Transformation:**



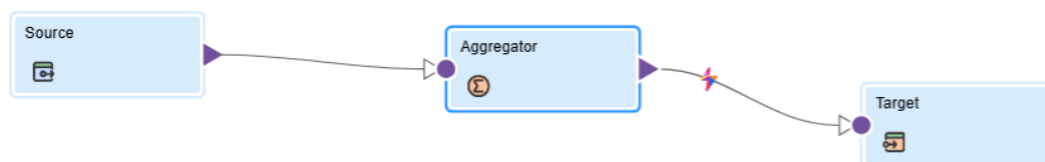
## Scenario 11: High-Level Analytics and Aggregation (m\_Agg\_Airline\_Delays)

- **Business Need:** Executives do not want to read millions of individual flight records. They need a quick summary showing the average delay time for each airline.
- **Transformation Logic:** An **Aggregator** transformation groups all the records by airline. It applies the AVG() math function to the departure delays, compressing thousands of rows into a lightweight, readable summary.
- **Transformation:**



## Scenario 12: Executive Summary Dashboard Rollup (m\_Executive\_Summary\_Dashboard)

- **Business Need:** The CEO needs a pristine, final report showing the exact total flight volumes and delay minutes for every airline, with absolutely no leftover data.
- **Transformation Logic:** An **Aggregator** rolls up millions of rows, using COUNT(), SUM(), and AVG() to calculate the final metrics. To guarantee the report is perfect, strict Field Rules are set to "Named Fields," blocking any unnecessary background columns from sneaking into the final four-column dashboard.
- **Transformation:**



Following the successful execution of the ETL pipelines in Informatica IDMC, the cleansed and aggregated data was loaded into Microsoft Power BI. An interactive Master Operations Dashboard was engineered to allow end-users to dynamically drill down into root causes for delays, track individual aircraft geographic routes, and monitor the high-level business metrics defined in the initial project objectives.



## 6. Conclusion

This project successfully demonstrated the complete, end-to-end design and implementation of an Aviation Data Warehouse, utilizing Informatica Intelligent Data Management Cloud (IDMC) for backend data engineering and Microsoft Power BI for front-end business intelligence.

Large-scale aviation datasets, consisting of flight, airline, and airport information, were seamlessly integrated through a series of complex ETL pipelines. Various Informatica transformations—including Router, Normalizer, Aggregator, Rank, and Sequence Generator—were deployed to address real-world data challenges such as data cleansing, deduplication, surrogate key generation, and outlier detection. This refined data was strategically structured into a Star Schema architecture, optimizing it for high-performance analytical processing.

Building upon this robust database foundation, the finalized data was ingested into Power BI to engineer a dynamic "Master Operations Dashboard." This enterprise-grade visualization tool successfully translated millions of rows of raw data into actionable insights, featuring interactive geographic flight tracking, automated root-cause delay analysis, and high-level executive KPI rollups.

Ultimately, this project highlights the entire data lifecycle—from extracting messy, unstructured CSV files to delivering pristine, decision-ready visual analytics. It proves that combining scalable data integration pipelines with professional UI/UX data visualization provides a powerful, comprehensive solution for operational monitoring in the modern aviation industry.